

Paper 4.50 - Boundary Repair Claim Contract

CQE-paper-04.50

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-04.50.md

Paper 4.50 - Boundary Repair Claim Contract

Purpose

Paper 4.50 defines what counts as a valid boundary repair claim. It prevents failed joins from being erased, and it prevents repair rows from being counted as proof before the next legal route closes them.

Admitted Paper 4 Claims

The following claims are admitted from Paper 4:

```
Paper 2 correction states can become typed boundary constraints
Paper 3 coordinates are preserved in the repair rows
required repair fields are state, axis_sheet, reason, status, next_legal_routes, source_paper,
target_paper
repaired rows have status constraint, not proof
repair is idempotent on already repaired rows
untyped failures are rejected
```

Claim Requirements

Any later paper using a Paper 4 repair row must state:

```
which correction state is being repaired
which axis/sheet coordinate is preserved
why the row exists
which next route consumes it
which receipt proves the row
whether the row remains a constraint or has been closed by a later theorem
```

Linked Receipt

The minimum receipt link for Paper 4 is:

```
paper: CQE-paper-04
theorem: Typed boundary repair
receipt: production/formal-papers/CQE-paper-04/boundary_repair_receipt.json
status: pass
```

The receipt is sufficient for the typed repair theorem. It is not sufficient by itself for oloid midpoint physics, confinement-scale claims, or downstream mass-residue claims.

Boundary Failures

The following are boundary failures:

```
using a row with status failed as if it were repaired
omitting the source state or axis/sheet coordinate
omitting the next legal route
marking a repair constraint as proof
allowing a payload to rewrite the repair row without a new receipt
```

Boundary failures are routed to obligations or rejected by the consuming paper.

Hidden-Guess Contract

When hidden-guess diagnostics are enabled, the answer must be recorded before the receipt is revealed. The guess record should include:

```
prompt
predicted required fields or repair status
revealed receipt
match/mismatch
next obligation if mismatch
```

The "constraint or proof?" prompt is required because the correct answer is constraint.

Conclusion

Paper 4.50 lets later papers import repair rows honestly. It keeps the failure visible, the repair typed, and the proof status bounded until another theorem actually closes the route.